DIMO DESIGN TOOLS · LIGHTING

Light Calc

Sports floodlighting designed, checked and reported in one place – on masts or a stadium roof, for one sport or several, from the first sketch to the approved report.



28 s
to a passing Class I football design

7 sports
plus several on one site

20
validation test groups, incl. CIE 171

1 button
place · size · aim · optimise · trim

1 **Sport & standard**
EN 12193, FIFA, broadcast

2 **Ground**
standard, OpenStreetMap or your drawing

3 **Luminaires**
maker's IES / LDT

4 **Masts or roof**
within the mast rules

5 **Design**
automatic, scene by scene

6 **Report**
PDF, aiming, .rdf, DXF

Designs, not only calculates

Places the masts, sets heights and floodlight numbers, aims and optimises every floodlight, then trims while every check still passes – for each luminaire and the mix, keeping the best.

Scenes, groups and DMX

Match, training, televised and security scenes, each checked against its own class, with its lowest passing level and the maximum demand for the supply.

TV cameras and glare

Standard camera positions, Ecam and uniformity per camera, glare to CIE 112 – and camera values for the broadcaster from the present lighting.

Stadium roof and roof ring

Stands, grass banks and eight roof types; floodlights on the catwalk, on masts, or both – roofs shade the masts' light exactly.

Neighbours, stands, colour

Obtrusive light to CIE 150 / EN 12464-2 before and after curfew; spectator light and glare; Ra, TLCI and flicker for UHD and slow-motion.

Photo-real view

Day, dusk or night, from the main camera or the pitch side – the pitch lit by the calculated lux. Save the picture for the proposal.

CASE STUDY · MULTI-SPORT

Athletics and football, roofed long sides, houses 215 m away

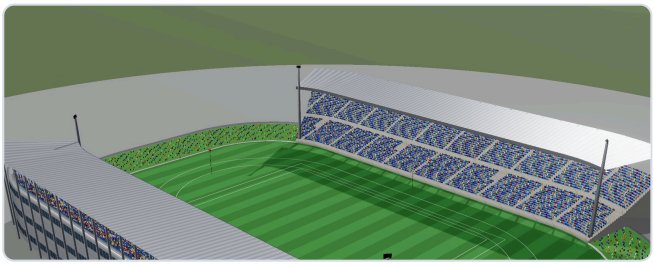
48
floodlights, masts + roof

72 kW
all on

4 / 4
scenes pass

88 s
automatic design

Scene levels set automatically (training at 20–35 %); neighbours' windows over the E3 limits found, with the fixes; camera values reported for the broadcaster instead of a must-pass TV scene that would have needed 942 kW.



- **PDF report** – standards, results, every scene, head frames with DMX groups, checks
- **Aiming PDF** for the installer; **point values** for the approver
- **Relux .rdf** in and out, **DXF** plan, **send to electrical** and high-mast foundations

Sports: cricket, football, rugby, athletics, tennis, pickleball, volleyball, multi-sport. **Standards:** EN 12193 Classes I–III, FIFA Classes I–V, international broadcast; obtrusive light EN 12464-2 / CIE 150; glare CIE 112. Results are point-by-point from the maker's photometry – confirm the final design in DIALux or Relux where required.